

Education

- Sep 2022 - **Ho Chi Minh University of Science, Vietnam National University**, Ho Chi Minh City.
Present Bachelor of Information Technology, Majoring in Knowledge Engineering (NLP)
GPA: 3.94/4.0 (9.29/10.0), Math: 4.0/4.0 (9.77/10.0)
- Aug 2019 - **Le Hong Phong high school for the gifted**, Ho Chi Minh City.
May 2022 Specialized Mathematics Class.

Experience

- Mar 2024 - **Research Student, AI-Thena, FIT-HCMUS.**
Present - Supervisors: Dr. Tung Le, Dr. Huy Tien Nguyen.
- Conducted research and experiments in Multimodal Processing, Fact Checking, Large Language Models, and advanced retrieval systems.
- Past research explorations include cross-lingual representation learning and knowledge distillation techniques to enable English-pretrained VLMs on Vietnamese VQA tasks.

Publications

- 2026 **MARC: Agent-Based Adaptive Routing for Multimodal Fact Verification.**
Huy Trieu, Duy-Minh Nguyen Tran, Huy Tien Nguyen, Tung Le
- Accepted at **KES 2026** (CORE B).
- Designed an adaptive multi-agent routing framework that dynamically delegates verification tasks to external retrieval or internal reasoning.
- Optimized context utilization, enabling compact vision-language models to achieve highly competitive performance against larger foundation models.
- 2026 **Improving Automated Essay Scoring with Targeted LLM-Based Data Augmentation under Imbalanced Data.**
Duy Anh Le, Nghia Vo Thanh, Huy Trieu, Van Chi Nam, Huy Tien Nguyen, Tung Le
- Accepted at **KES 2026** (CORE B).
- Proposed a novel band-aware LLM-driven data synthesis framework that dynamically profiles prediction errors and generates targeted training samples for underrepresented IELTS proficiency bands.
- 2025 **KWordinaryVQA: A Keyword-Driven Generative Visual Question Answering System for Culinary Exploration.**
Huy Trieu, Thanh Thai Nguyen, Thanh Nghia Vo, Thinh Vuong Vo, Thanh Tu Dang, Tung Le
- Published in **PACLIC 39** (CORE C).
- Designed and implemented an automated end-to-end pipeline for large-scale food-domain VQA data generation.
- Publicly released KWordinaryVQA dataset (43,455 QA pairs, 8,693 images) as a new benchmark for VLM reasoning.
- Conducted extensive benchmarking of retrieval, classification, and generative (LLMs) approaches.
- [Paper](#)
- 2025 **Flame Reavers@ALQAC 2025: Integrating Learned Rankers and LLM Reasoning in a Dynamic Hybrid Architecture for Legal Retrieval.**
Huy Trieu, Dang-Phuong-Nam Doan, Anh-Kiet Nguyen, Thanh-Thai Nguyen, Thanh-Nghia Vo, Tung Le, Huy Tien Nguyen
- Published in **KSE 2025** (CORE N/A, Q3 Scopus).
- Developed a dynamic hybrid retrieval system combining sparse/dense retrievers, pairwise learning-to-rank, hard-negative-mined Cross-Encoder, and adaptive LLM reranking.
- Achieved 1st place in Legal Document Retrieval task and 5th place in Legal QA task.
- [Paper](#)

Achievements & Honors

- 2022–2026 **University Award, Academic Merit Scholarship**, Faculty of Information Technology.
Awarded for outstanding academic performance, ranking in the top 5% of IT students across 6 semesters.
- 2025 **ALQAC 2025 Competition**, Flame Reavers, Leader.
First Place in Legal Document Retrieval task with a decisive lead in performance metrics over remaining competitors.
- 2023–2024 **University Award, Dean’s List**, Faculty of Information Technology.
Maintained excellent academic standing, ranking in the top 1% of IT students in an academic year.
- 2021–2022 **High School Competition.**
Third place in the mathematics competition for 12th grade at the city level.
- 2018–2019 **Middle School Competition.**
Second place in the mathematics competition using calculators for 9th grade at the city level.